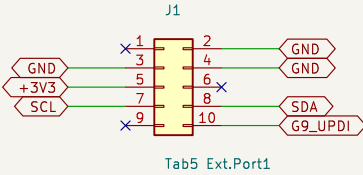
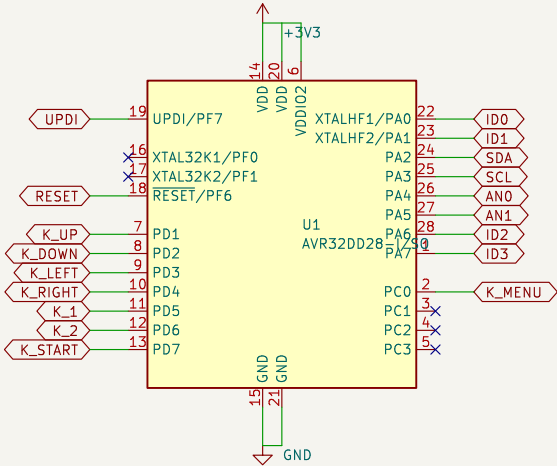
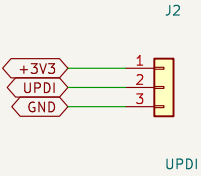


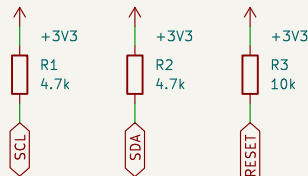
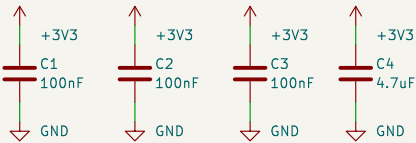
MCU – AVR32DD28 (SOIC–28) running firmware\_avrdd (I2C client 0x6D, RetroPad Ext.Port1) – verify pin 1 orientation before fab  
Buttons – one MCU pin each, switch to GND (internal pull–up); slot table in firmware\_avrdd/pinmap.h  
PD1–PD7, PC0–PC3, PF0–PF1: one button per pin (internal pull–ups)



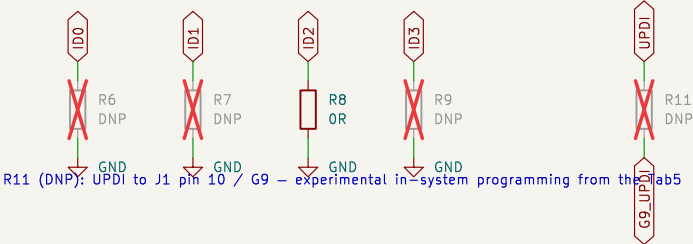
UPDI programming (SerialUPDI / MPLAB Snap)



Decoupling (VDD x2, VDDIO2), I2C pull–ups, RESET pull–up

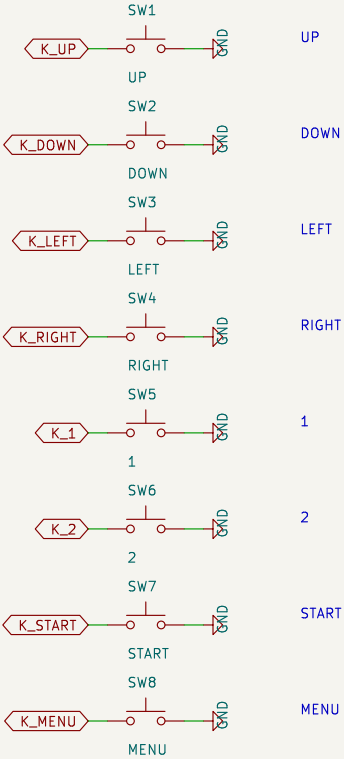
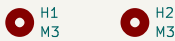


Console ID straps: OR to GND = bit set. Master System / Game Gear = ID 4 (fit R8; R6, R7, R9 DNP)



R11 (DNP): UPDI to J1 pin 10 / G9 – experimental in–system programming from the Tab5

Mounting holes (M3, 96 mm apart)



Tab5 keyboard–port board: AVR32DD28 + firmware\_avrdd, I2C 0x6D  
Generated by kicad\_gen/gen\_sch.py from gen\_pcb.py and layout.py  
**RetroESP32–P4**  
Sheet: /  
File: retropad\_sms.kicad\_sch

**Title: RetroPad Master System / Game Gear controller (console ID 4)**

Size: A3 Date: 2026–10–05 Rev: 0.1  
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4 Id: 1/1